

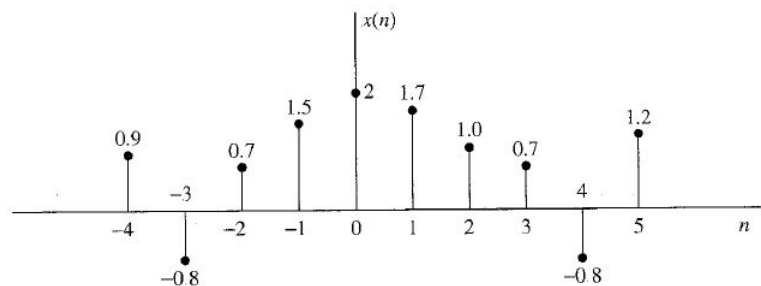
CSE 4631

Digital Signal Processing

Week 2 & 3

Representation of Discrete-Time Signals

Graphical Representation



Functional Representation

$$x(n) = \begin{cases} 1, & \text{for } n = 1, 3 \\ 4, & \text{for } n = 2 \\ 0, & \text{elsewhere} \end{cases}$$

Tabular Representation

n	...	-2	-1	0	1	2	3	4	5	...
$x(n)$	...	0	0	0	1	4	1	0	0	...

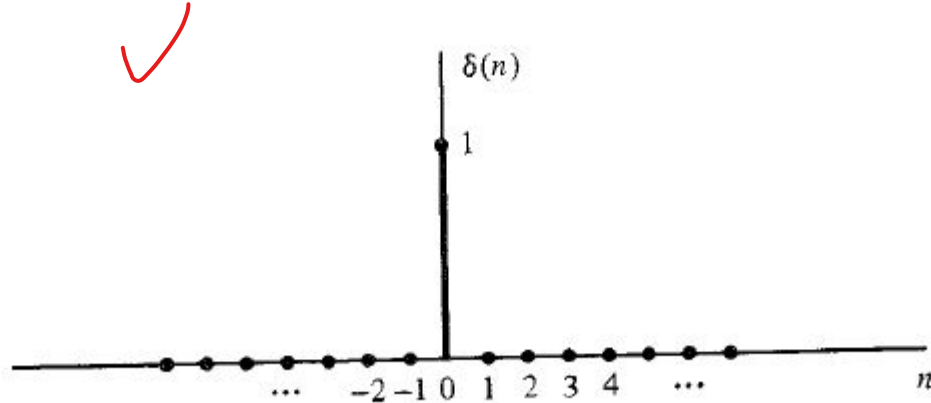
Sequence Representation

$$x(n) = \{ \dots 0, 0, 1, 4, 1, 0, 0, \dots \}$$

↑

Some Elementary Discrete-Time Signals

- **Unit sample sequence:** Zero everywhere, except at $n = 0$ where its value is unity. Also referred to as a ~~unit impulse~~.



$$\delta(n) \equiv \begin{cases} 1, & \text{for } n = 0 \\ 0, & \text{for } n \neq 0 \end{cases}$$

Figure: Graphical representation of the unit sample signal

Some Elementary Discrete-Time Signals (contd.)

- Unit step signal

$$u(n) \equiv \begin{cases} 1, & \text{for } n \geq 0 \\ 0, & \text{for } n < 0 \end{cases}$$

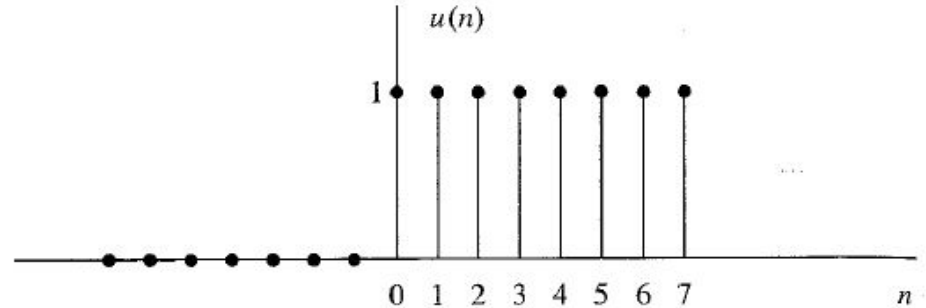


Figure: Graphical representation of the unit step signal

- Unit ramp signal

$$u_r(n) \equiv \begin{cases} n, & \text{for } n \geq 0 \\ 0, & \text{for } n < 0 \end{cases}$$

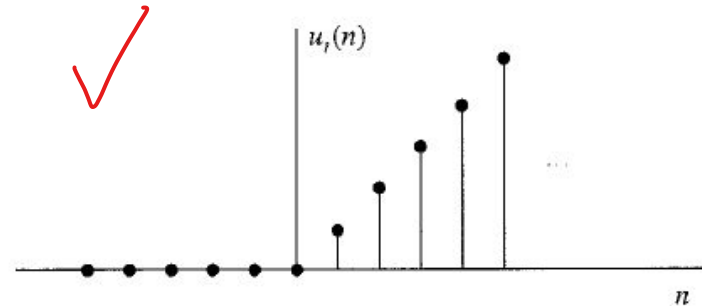


Figure: Graphical representation of the unit ramp signal

Some Elementary Discrete-Time Signals (contd.)

- Exponential signals

$$x(n) = a^n \quad \text{for all } n$$

$$x(n) = r^n e^{j\theta n}$$

$$= r^n (\cos \theta n + j \sin \theta n)$$

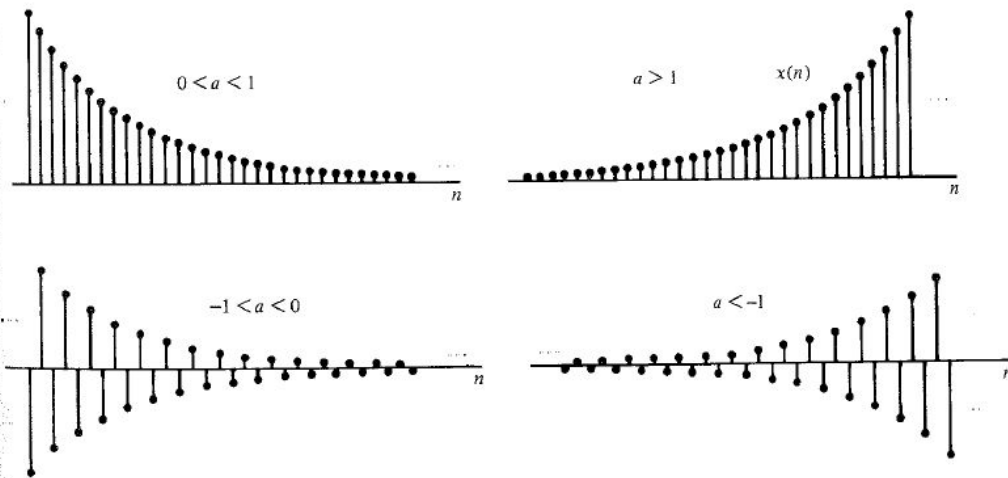
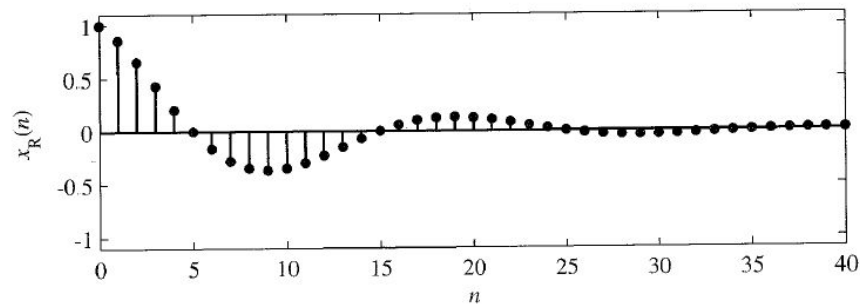
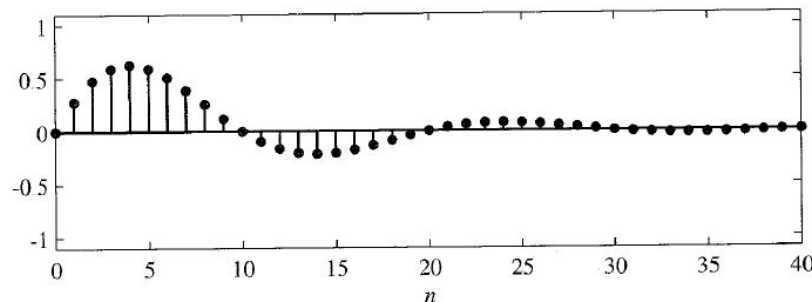


Figure: Graphical representation of a real-valued exponential signal



(a)



(b)

Figure: Graphical representation of the (a) real and (b) imaginary components of a complex-valued exponential signal

Classification of Discrete-Time Signals

- **Energy signals and power signals**

- If E is finite ($0 < E < \infty$) then $x(n)$ is called an **energy signal**. energy signals have zero average power over infinite time.
- If P is finite and non-zero then $x(n)$ is called a **power signal**. power signals have infinite total energy.
- A signal cannot be both power signal and energy signal.

$$\left[E \equiv \sum_{n=-\infty}^{\infty} |x(n)|^2 \right]$$

$$\left[P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 \right]$$

Question: Determine if the unit step sequence is a power signal or an energy signal.

Classification of Discrete-Time Signals (contd.)

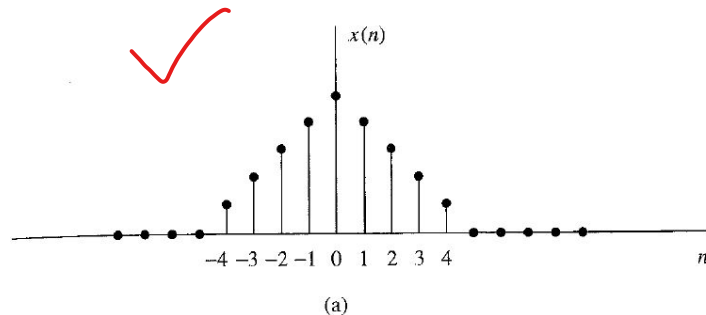
- **Periodic signals and aperiodic signals**

- A signal $x(n)$ is periodic with period N ($N > 0$) if and only if $x(n+N) = x(n)$ for all n
- The smallest value of N for which a signal is periodic is called the **fundamental period**
- If there is no value of N that satisfies this property, the signal is called ~~nonperiodic or aperiodic~~ 
- **Periodic signals are power signals.** Their average power is the same across all periods.

Classification of Discrete-Time Signals (contd.)

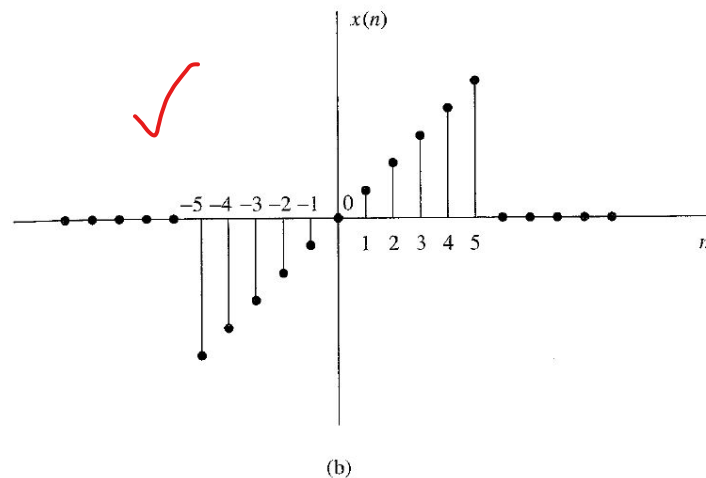
- Symmetric (even) signals

$$x(-n) = x(n)$$



- Antisymmetric (odd) signals

$$x(-n) = -x(n)$$



Note: If $x(n)$ is odd, then $x(0) = 0$

Figure: Example of even (a) and odd (b) signals

Any arbitrary signal can be expressed as the sum of two signal components, one of which is *even* and the other is *odd*.

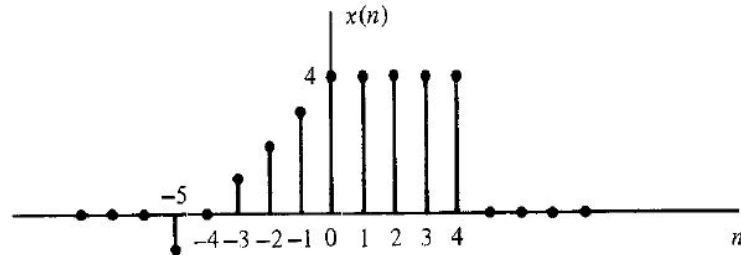
$$\left[x_e(n) = \frac{1}{2} [x(n) + x(-n)] \right]$$

$$\left[x_o(n) = \frac{1}{2} [x(n) - x(-n)] \right]$$

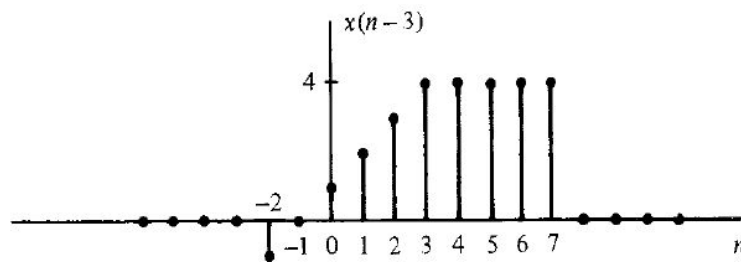
$$\underline{x(n) = x_e(n) + x_o(n)}$$

Simple Manipulations of Discrete-Time Signals

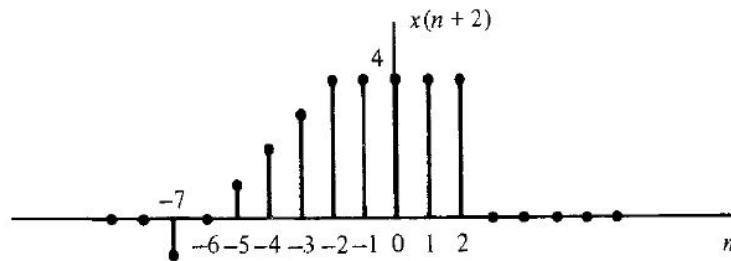
- Transformation of the independent variable (time)
 - $x(n)$ may be shifted in time by replacing n by $n-k$ where k is an integer.
 - If k is positive, the time shift results in a delay of the signal by k units of time.
 - If k is negative, the time shift results in an advance of the signal by $|k|$ units in time.
 - ~~○ If the signal is stored then it is relatively simple operation to modify the base by introducing a delay or an advance.~~
 - ~~○ However, it is impossible to advance a signal that is being generated in real time.~~



(a)



(b)



(c)

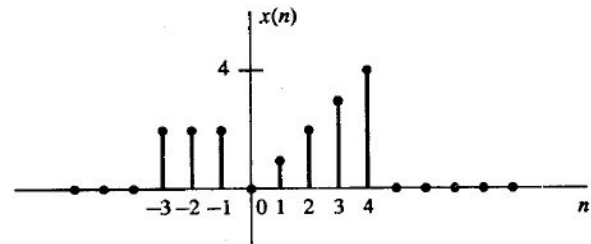
Figure: Graphical representation of a signal and its delayed and advanced versions

Simple Manipulations of Discrete-Time Signals (contd.)

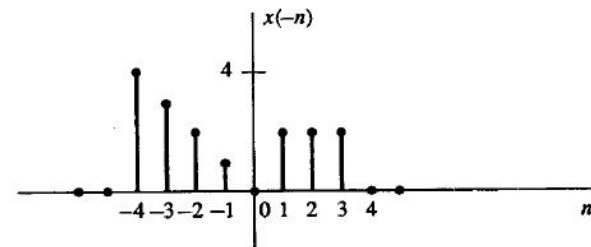
- Transformation of the independent variable (time) (contd.)
 - Another useful modification of the time base is to **replace the independent variable n by $-n$**
 - The result of this operation is **a folding** or a **reflection of the signal about the time origin $n=0$**
 - The **operations of folding and time delaying or advancing are not commutative.**

$$\text{TD}_k\{\text{FD}[x(n)]\} = \text{TD}_k[x(-n)] = x(-n + k)$$

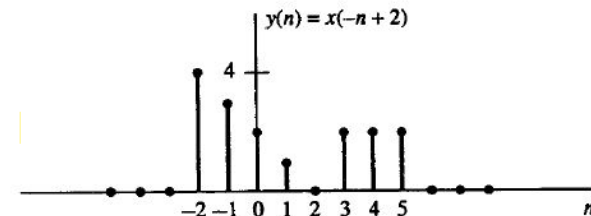
$$\text{FD}\{\text{TD}_k[x(n)]\} = \text{FD}[x(n - k)] = x(-n - k)$$



(a)



(b)



(c)

Figure: Graphical representation of the folding and shifting operations

Simple Manipulations of Discrete-Time Signals (contd.)

- Down-sampling

- $y(n) = x(kn)$
- Sampling interval increases to kT (From T)
- Sampling rate drops to $1/kT$ (From $1/T$)

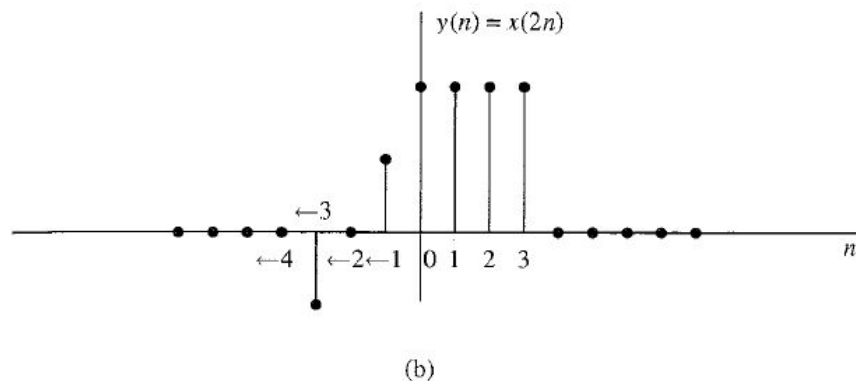
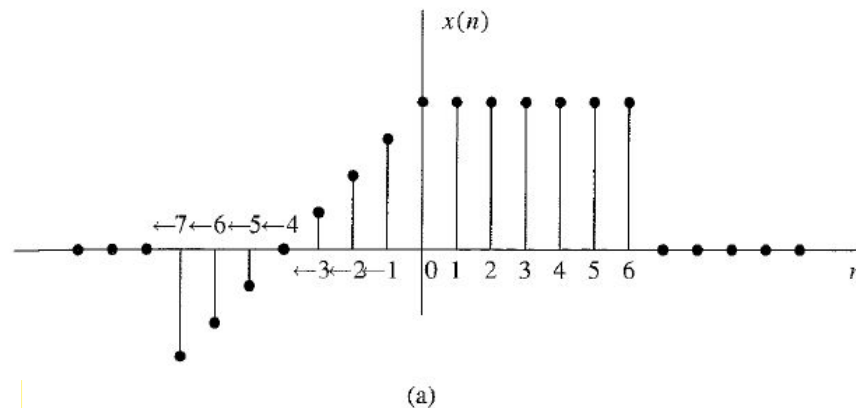
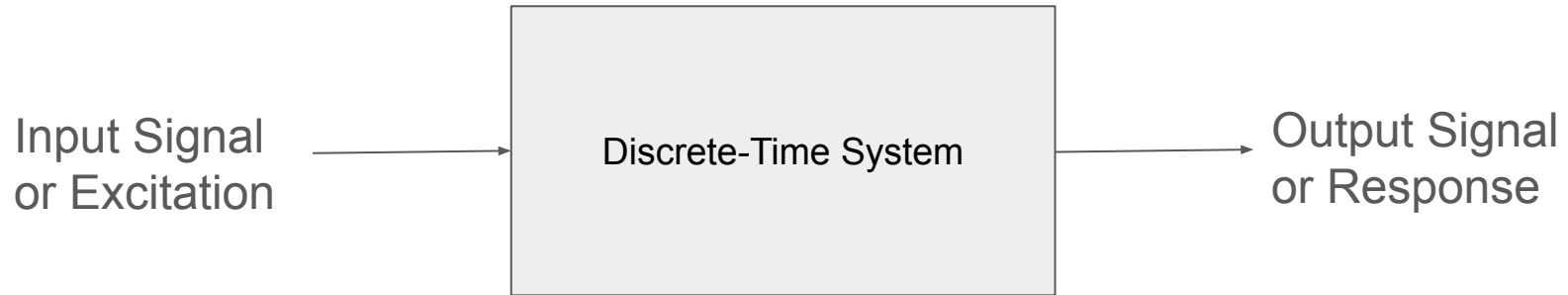


Figure: Graphical illustration of down-sampling operation

Simple Manipulations of Discrete-Time Signals (contd.)

- **Addition:** $y(n) = x_1(n) + x_2(n), \quad -\infty < n < \infty$
- **Multiplication:** $y(n) = x_1(n) x_2(n), \quad -\infty < n < \infty$
- **Amplitude scaling:** $y(n) = Ax(n), \quad -\infty < n < \infty$

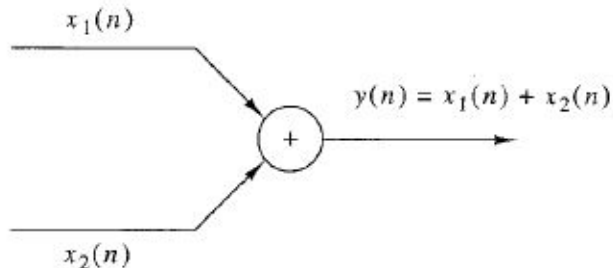
Discrete-Time Systems



Block diagram Representation of Discrete-Time Systems

- Adder

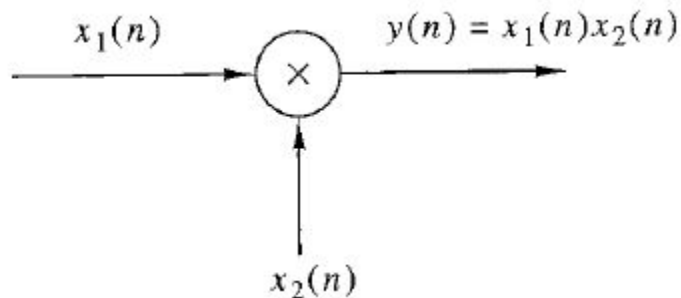
- Memoryless



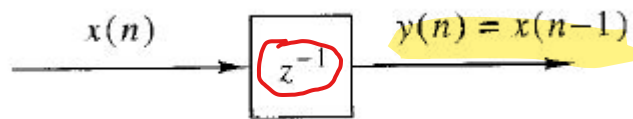
- Constant Multiplier



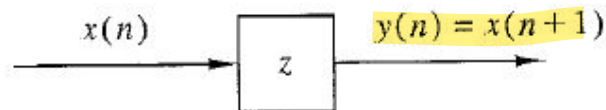
- Signal Multiplier



- A unit delay element

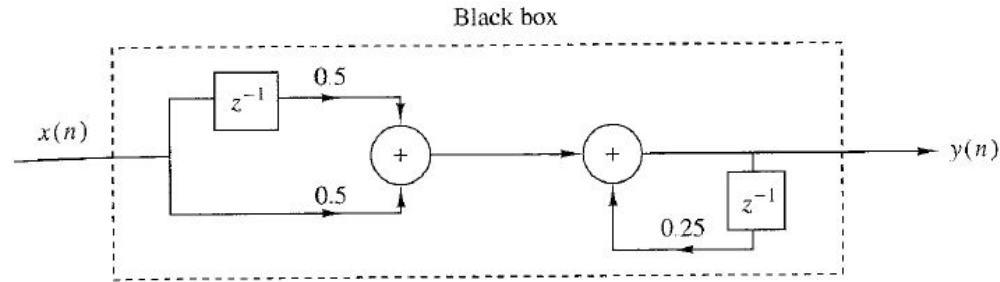


- A unit advance element

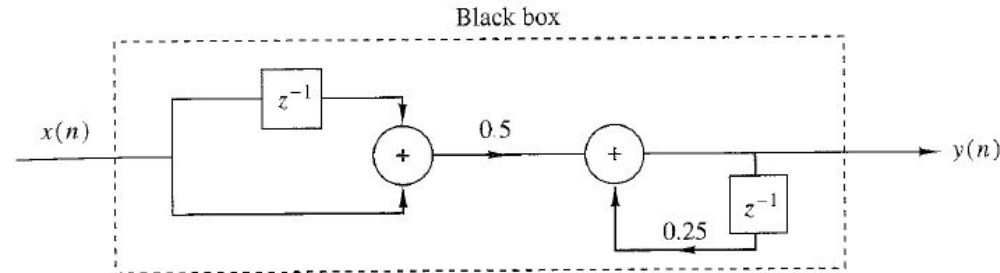


Block diagram Representation of Discrete-Time Systems (Contd.)

$$\left[y(n) = \frac{1}{4}y(n-1) + \frac{1}{2}x(n) + \frac{1}{2}x(n-1] \right]$$



(a)



(b)

Classification of Discrete-Time Systems

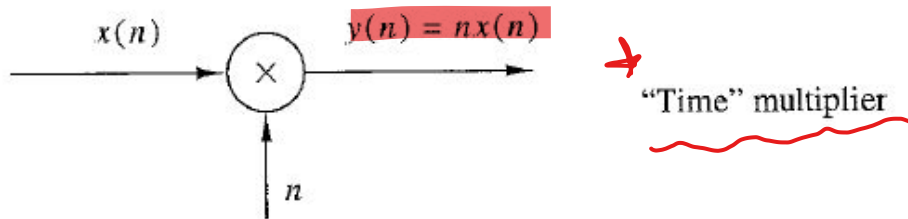
- **Static (memoryless) systems:** Output at any instant n depends on at most on the input sample at the same time.
- **Dynamic systems:** Output can depend on past or future samples of the input.
 - If the output at time n is completely determined by the input samples in the interval from $n - N$ to n ($N \geq 0$), the system is said to have memory of duration N .
 - If $N = 0$, the system is static.
 - If $0 < N < \infty$, the system has finite memory
 - If $N = \infty$, the system has infinite memory

Classification of Discrete-Time Systems (contd.)

- **Time-invariant systems**

- Input-output characteristics do not change with time
- A system $\mathcal{T}$ is *time invariant* or *shift invariant* if and only if $x(n) \xrightarrow{\mathcal{T}} y(n)$ implies that $x(n-k) \xrightarrow{\mathcal{T}} y(n-k)$

- **Time-variant systems**

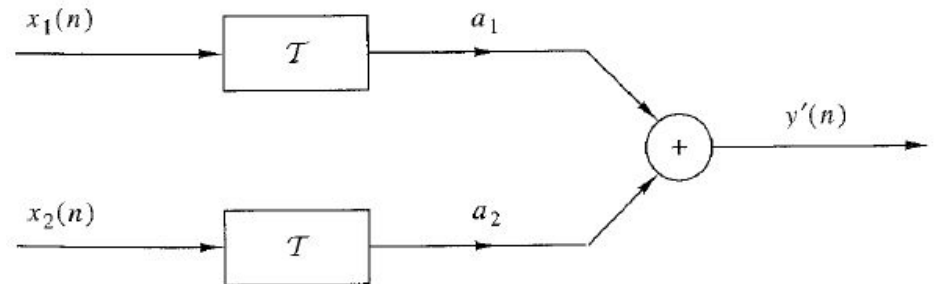
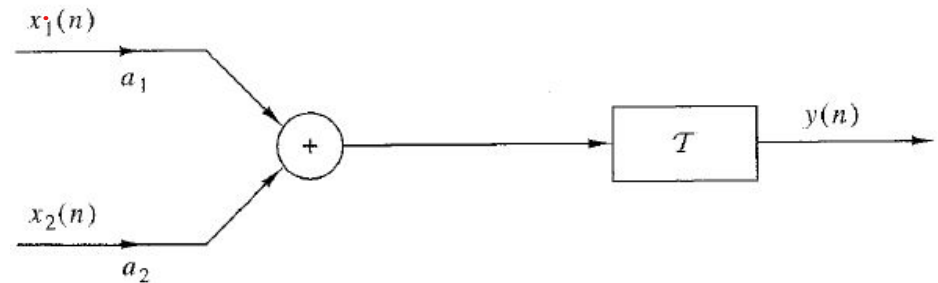


Classification of Discrete-Time Systems (contd.)

- **Linear versus nonlinear systems**

- Satisfies the **superposition principle**
- Satisfies the **multiplicative/scaling principle** (If, suppose, $a_2 = 0$)
- Satisfies the **additivity principle** (If, suppose, $a_1 = a_2 = 1$)

$$\mathcal{T}[a_1x_1(n) + a_2x_2(n)] = a_1\mathcal{T}[x_1(n)] + a_2\mathcal{T}[x_2(n)]$$



Classification of Discrete-Time Systems (contd.)

Find out the linearity of the following systems described by these input-output equations:

- $y(n) = nx(n)$
- $y(n) = Ax(n) + B$

Classification of Discrete-Time Systems (contd.)

• Causal systems

- Output depends only on present and past inputs but not on future inputs
- Satisfies the following function where $F[\]$ is an arbitrary function:

$$y(n) = F[x(n), x(n-1), x(n-2), \dots]$$

• Noncausal systems

- Output depends also on future inputs
- Physically unrealizable for real-time signal processing applications
- Possible to implement if the processing is done offline with recorded signal (e.g., geophysical signals and images)

Classification of Discrete-Time Systems (contd.)

Which of the following systems described by these input-output equations are causal and which ones are noncausal?

- $y(n) = x(n) - x(n-1)$
- $y(n) = x(n) + 3x(n+4)$
- $y(n) = x(n^2)$
- $y(n) = ax(n)$
- $y(n) = x(2n)$
- $y(n) = x(-n)$

Classification of Discrete-Time Systems (contd.)

- **Stable versus unstable systems**

- Every bounded input produces a bounded output
- Also called bounded input-bounded output (BIBO) stable
- There exist some finite numbers, M_x and M_y , such that

$$\underline{|x(n)| \leq M_x < \infty, \quad |y(n)| \leq M_y < \infty}$$

- / Show that the system described by the following input-output equation is unstable:

$$y(n) = y^2(n-1) + x(n)$$

Interconnection of Discrete-Time Systems

- **Cascade (series) interconnection**

$$y_1(n) = \mathcal{T}_1[x(n)]$$

$$y(n) = \mathcal{T}_2[y_1(n)]$$

$$= \mathcal{T}_2\{\mathcal{T}_1[x(n)]\}$$

$$\underline{\mathcal{T}_c \equiv \mathcal{T}_2 \mathcal{T}_1}$$

$$\underline{y(n) = \mathcal{T}_c[x(n)]}$$

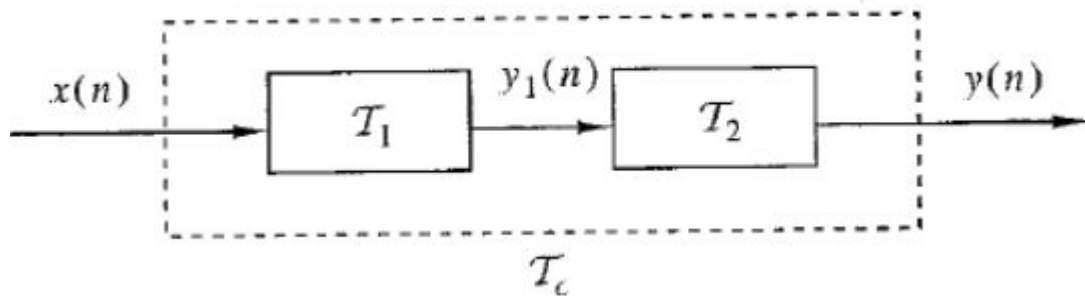


Figure: Cascade interconnection of systems

In general, $\mathcal{T}_2 \mathcal{T}_1 \neq \mathcal{T}_1 \mathcal{T}_2$

However, if the systems are linear and time invariant, then $\mathcal{T}_c$ is time invariant and $\mathcal{T}_2 \mathcal{T}_1 = \mathcal{T}_1 \mathcal{T}_2$,

Interconnection of Discrete-Time Systems (contd.)

- Parallel interconnection

$$\begin{aligned}y_3(n) &= y_1(n) + y_2(n) \\&= \mathcal{T}_1[x(n)] + \mathcal{T}_2[x(n)] \\&= (\mathcal{T}_1 + \mathcal{T}_2)[x(n)] \\&= \mathcal{T}_p[x(n)]\end{aligned}$$

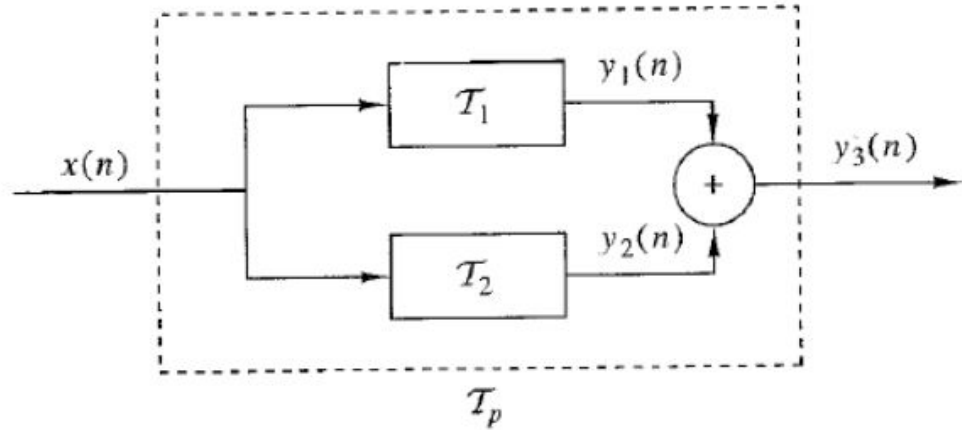


Figure: Parallel interconnection of systems